
Cost-Effectiveness of CRT vs TORS for HPV- Associated Oropharyngeal Cancer



Objective of the Presentation

1

Evaluate Sher et al. (2016) cost-effectiveness study

2

Explain CEA principles for non-economists

3

Determine if TORS investment is justified

4

Provide clear recommendation for leadership

What is Cost-Effectiveness Analysis (CEA)?

Compares costs and health outcomes of interventions

Outputs: **QALY** (Quality-Adjusted Life Year)

Key metric: **ICER** (Incremental Cost-Effectiveness Ratio)

Benchmarks:

< \$50,000/QALY = **Very cost-effective**

\$50–100K = **Acceptable**

\$100K = **Less cost-effective**

Study Being Evaluated

Study: Sher et al., 2016

Population: 65-year-old male, HPV-positive, N2 oropharyngeal cancer

Strategies Compared:

CRT (definitive chemoradiation)

TORS + adjuvant RT or CRT (if needed)

Used **Markov model**, 10-year time horizon, payer perspective

Understanding the Markov Model

Health states: NED
(no evidence of
disease),
recurrence,
metastasis, death

Transitions based
on probabilities

Tracks outcomes
monthly over 10
years

Includes utilities
and cost per state

Base Case Results

Treatment	QALYs	Cost (USD)	Outcome
CRT	7.31	\$50,100	Dominant
TORS	7.29	\$62,200	More costly, less effective

What Drives TORS Costs?

01

61.2% of
TORS patients
required
adjuvant CRT

02

PEG feeding
needed in
TORS + CRT
cases

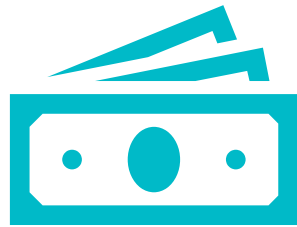
03

Equipment
and surgical
costs: \$16,000
–\$29,000

04

Complication
s can increase
cost to
\$25,895+

Sensitivity Analysis Summary



**CRT remains cost-effective
under most scenarios**



TORS only cost-effective if:

Recurrence rate drops **>30%**
Long-term utility is **≥ 0.03 higher**
No adjuvant CRT required

Strengths of the Study



REALISTIC DATA
INPUTS (CLINICAL
TRIALS, MEDICARE
RATES)



ROBUST MODELING (1-
WAY, 2-WAY,
PROBABILISTIC)



INCLUDES TOXICITY
AND UTILITY IMPACTS



VALUE OF
INFORMATION
ANALYSIS INCLUDED

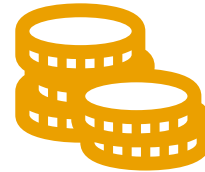
Limitations of the Study



Assumes **equal long-term quality of life**



High rate of adjuvant CRT (61%) may be outdated



2015 cost data — may underestimate TORS improvements



Focused on one patient profile (non-smoker, age 65)

Perspective and Implications

1

Payer perspective:
focuses on direct treatment costs

2

Societal perspective: may favor TORS for faster recovery, less caregiver time

3

Hospital leadership must weigh **patient-centered care** vs **financial sustainability**

Real-World Implications

TORS could become viable if:

- Adjuvant CRT rates <30%
- Better functional outcomes proven
- Robotic costs decrease

Current evidence does **not justify** major TORS investment

Final Recommendation

-
- Do **not invest** in a TORS program yet
 - Continue CRT as standard of care
 - Reevaluate after ECOG 3311 trial data
 - Monitor real-world outcomes on recurrence, adjuvant needs, and utility

Key Takeaways

- CRT = 7.31 QALYs @ \$50,100 → more effective, cheaper
- TORS = 7.29 QALYs @ \$62,200 → costlier, minor benefit
- Cost-effectiveness depends on uncertain TORS outcomes
- CRT remains best value **for now**

Reference

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THANK YOU